

Phase 6 TSP Replay Mechanism Runbook

This runbook covers the focused mechanism follow-up after the first routing archive.

The scientific target is narrower than phase 5:

- explain why random_replay beat failure_replay on symmetric TSPLIB TSP
- separate broad replay coverage from raw failure severity
- test whether residual-failure replay and compression change that conclusion

1. Prepare The Benchmark Pack

Activate the environment first:

```
conda activate Python3_14
python prepare_tsp_benchmarks.py
```

That refreshes benchmarks/tsp/manifest.json with descriptor payloads for the official TSPLIB and synthetic TSP instances.

2. Offline Smoke Test

```
python run_tsp_suite.py --config configs/tsp_phase6_suite/smoke_suite.json --skip-judge
```

Use this first whenever you change descriptor extraction, residual-difficulty logic, replay selection, or the phase-6 aggregate code. The smoke suite writes into DO NOT COMMIT/tsp_phase6_suite/smoke.

3. Main Phase-6 Suite

```
python run_tsp_suite.py --config configs/tsp_phase6_suite/01_replay_mechanism.json
```

The nine official conditions are:

- phase6_no_replay
- phase6_random_replay
- phase6_failure_replay
- phase6_random_replay_compression
- phase6_stratified_random_replay
- phase6_diversity_weighted_replay
- phase6_residual_failure_replay
- phase6_diversity_failure_replay
- phase6_diversity_failure_replay_compression

Primary interpretation targets:

- final held-out TSPLIB optimality gap
- final synthetic holdout gap
- replay archive descriptor diversity
- replay archive hardness

Key secondary diagnostics:

- archive size bias
- replay failure concentration
- accepted code novelty
- heuristic complexity
- adaptation efficiency

4. Replication

Use paired seed offsets exactly as in earlier phases:

```
python run_tsp_suite.py --config configs/tsp_phase6_suite/01_replay_mechanism.json --seed-offset 1000  
--replicate-label b
```

Official target:

- 20 paired offsets using the fixed list 0, 1000, 2000, ..., 19000

5. Aggregation

```
python aggregate_tsp_phase6_runs.py --runs-root runs/tsp_phase6_suite/replay_mechanism
```

That writes:

- aggregate_summary.json
- aggregate_report.md
- aggregate_report.pdf
- archive_diversity_vs_tsplib_gap.svg/png
- archive_hardness_vs_tsplib_gap.svg/png

6. Interpretation Rules

- Treat this as a mechanism study, not as a new solver benchmark race.
- The main comparison is not only replay versus no replay. It is broad-coverage replay versus raw-failure replay versus residual-failure replay.
- If archive diversity predicts held-out transfer more strongly than archive hardness, say that directly and scope the claim to this TSP benchmark family.
- random_replay_compression matters because it tests compression on the replay arm that actually won phase 5.
- Do not call residual replay a success unless it improves held-out transfer or clearly sharpens the diversity-versus-hardness explanation.